

Population-level morphological analysis of paired CO₂- and odor-sensing olfactory neurons in *D. melanogaster* via volume electron microscopy

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eLife Assessment

This **valuable** study reveals surprising morphological diversity of *Drosophila* sensory neurons. Using serial block-face electron microscopy, the authors created detailed 3D reconstructions of large neuronal populations, **convincingly** finding significant structural variation both within and across distinct classes. These results form the basis for testable hypotheses on how neuronal arborization is optimized for particular sensory functions. This research will be highly relevant to biologists in the fields of physiology, insect chemosensation, and neuroscience.

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Abstract

Dendritic morphology is a defining characteristic of neuronal subtypes. In *Drosophila*, heterotypic olfactory receptor neurons (ORNs) expressing different receptors display diverse dendritic morphologies, but whether such diversity exists among homotypic ORNs remains unclear. Using serial block-face scanning electron microscopy on cryofixed tissues, we analyzed the majority of CO₂-sensing neurons (ab1C) and their odor-sensing neighbors (ab1D) in the *D. melanogaster* antenna. Surprisingly, ab1C neurons featured flattened, sheet-like dendrites—distinct from the cylindrical branches typical of odor-sensing neurons—and displayed remarkable diversity, ranging from plain sheets to tube-like structures that enclose several neighboring dendrites, forming “dendrite-within-dendrite” structures. Similarly, ab1D dendrites varied from simple, unbranched forms to numerous branched morphologies. These findings suggest that morphological heterogeneity is common even among homotypic ORNs, potentially expanding their functional adaptability and ranges of sensory physiological properties.

Introduction

Dendrites are neuronal processes specialized to receive information. In observing the bewildering variety of dendritic morphologies, Ramón y Cajal famously postulated that “all the morphological features displayed by neurons appear to obey precise rules that are accompanied by useful consequences” (Cajal, 1995 [\[1\]](#)). What might these useful consequences be? Dendritic size and complexity vary to meet the functional demands of specific neurons (Hall & Treinin, 2011 [\[2\]](#); Jan & Jan, 2001 [\[3\]](#), 2010 [\[4\]](#)). For instance, the dendritic arborizations of somatosensory neurons define the geometry and size of their receptive fields (Hall & Treinin, 2011 [\[2\]](#)). On the other hand, the numerous or elongated dendrites of olfactory receptor neurons (ORNs) are thought to enhance sensory surface area for heightened sensitivity (Challis et al., 2015 [\[5\]](#); Smith, 2008 [\[6\]](#)). This is exemplified by the *Manduca* moth, whose long ORN dendrites in trichoid sensilla are correlated with the insect’s exquisite pheromone acuity (K.-E. Kaissling et al., 1989 [\[7\]](#); K. E. Kaissling, 1996 [\[8\]](#); Keil, 1989 [\[9\]](#); Lee & Strausfeld, 1990 [\[10\]](#)).

Dendritic morphologies are thus broadly used as defining features for specific neuronal subtypes (Jan & Jan, 2010 [\[4\]](#)). Indeed, in *D. melanogaster*, ORNs expressing different receptors or housed in distinct morphological types of sensilla exhibit diverse dendritic structures. For example, the outer dendrites of basiconic and intermediate ORNs are numerous branched, whereas those of coeloconic and trichoid neurons are typically unbranched (Nava Gonzales et al., 2021 [\[11\]](#); Shanbhag et al., 1999 [\[12\]](#)). Using serial block-face scanning electron microscopy (SBEM) and nanoscale morphometric analysis, we further found that the sensory surface areas of outer dendrites, where olfactory receptors are localized (Benton et al., 2006 [\[13\]](#), 2009 [\[14\]](#)), vary significantly among these morphological classes. ORNs with numerous branched dendrites generally have greater surface areas. Specifically, coeloconic neurons have the smallest sensory surface ($1\text{--}9\ \mu\text{m}^2$), followed by trichoid ($18\text{--}44\ \mu\text{m}^2$), intermediate ($11\text{--}47\ \mu\text{m}^2$), and basiconic ORNs, which exhibit the largest range ($11\text{--}275\ \mu\text{m}^2$) (Nava Gonzales et al., 2021 [\[11\]](#)). Interestingly, ORN types with extensive dendritic branches also exhibit larger, mitochondria-rich inner dendritic segments (Nava Gonzales et al., 2021 [\[11\]](#)), suggestive of high metabolic demands to sustain such structures. Therefore, diverse dendritic morphologies likely support specialized functions, with basiconic ORNs’ highly branched dendrites enhancing food odor sensitivity, while coeloconic ORNs’ short and unbranched dendrites reduce metabolic costs for detecting volatile acids or amines.

Does similar morphological diversity exist among homotypic ORNs expressing the same receptor? Addressing this question is crucial for understanding how structural variations within a single neuronal type may contribute to fine-tuning odor sensitivity or additional specialized functions. In our published SBEM study, we reconstructed 3D models for four Or67d-expressing trichoid ORNs from the same antenna. Two neurons had unbranched dendrites, while the others had three and six dendritic branches, respectively. Interesting, one unbranched dendrite displayed a cylindrical morphology proximally, transitioned to a slightly flattened structure mid-length, and reverted to a cylindrical form distally (Nava Gonzales et al., 2021 [\[11\]](#)). While these four neurons represent only a small subset of the 55–60 Or67d neurons in an antenna (Grabe et al., 2016 [\[15\]](#); Shanbhag et al., 1999 [\[12\]](#)), these observations suggest that dendritic morphology among homotypic neurons may be less uniform than previously assumed. However, determining the full extent of dendritic heterogeneity within an identified ORN population requires a systematic morphological and morphometric analysis at the population level.

To this end, here we generated a new SBEM volume encompassing nearly the entire antenna of *D. melanogaster*. The tissues were processed using the CryoChem method, which we previously developed to achieve high-quality ultrastructural preservation of cryofixed samples for volume EM (Tsang et al., 2018 [\[16\]](#)). Cryofixation via high pressure freezing and freeze-substitution is particularly crucial for properly preserving tissues with cuticles (such as insect sensory

appendages) that are impermeable to chemical fixatives. Additionally, cryofixation offers superior morphological preservation to then allow for more accurate morphometric measurements, unlike chemical fixation which can distort membrane structures, thus precluding faithful quantification (Shanbhag et al., 1999 [↗](#), 2000 [↗](#), 2001 [↗](#); Tsang et al., 2018 [↗](#)).

Furthermore, through genetic labeling of multiple identified ORNs in SBEM volumes, our prior studies established that within a sensillum, the rank order of ORN sizes corresponds to the neurons' relative extracellular spike amplitudes, designated as “A”, “B”, “C”, or “D” ORNs in descending order of spike size (Nava Gonzales et al., 2021 [↗](#); Tsang et al., 2018 [↗](#); Zhang et al., 2019 [↗](#)). Therefore, once the sensillum identity is confirmed, the cellular identities of ORNs can be inferred based on their relative neuronal sizes, without requiring genetic labeling. Moreover, once the cellular identity of an ORN is established, its corresponding olfactory receptor can be determined based on published molecular and functional atlases of fly ORNs (Benton et al., 2009 [↗](#), 2025 [↗](#); Couto et al., 2005 [↗](#); Fishilevich & Vosshall, 2005 [↗](#); Hallem et al., 2004 [↗](#); Task et al., 2022 [↗](#); Yao et al., 2005 [↗](#)).

Using these established and validated approaches, we characterized the 3D structures and measured the nanoscale morphometrics of CO₂-sensing ab1C neurons and their odor-sensing ab1D neighbor in *D. melanogaster*. CO₂ is a particularly ethologically significant odorant, as it serves as an alarm signal emitted from fruit flies (Suh et al., 2004 [↗](#)). Additionally, the nanoscale features of these neurons have not been previously characterized. Of note, ab1C and ab1D neurons reside within the ab1 sensillum, the most abundant basiconic sensillum type on the antenna (Grabe et al., 2016 [↗](#)), making it an ideal model for examining homotypic neurons' morphologies at a population-level.

In this study, we reconstructed 3D models for over 50% of the ab1C and ab1D neurons within a single antenna, enabling a broad-scale survey of morphological variability. Notably, ab1C neurons exhibited flattened, sheet-like dendrites, in stark contrast to the cylindrical branches typical of odor-sensing neurons. Both ab1C and ab1D dendrites displayed remarkable diversity, suggesting that morphological heterogeneity is a common feature among homotypic ORNs. Our findings suggest that while homotypic ORNs exhibit characteristic dendritic motifs, their morphologies are more variable than Cajal's “precise rules” might imply.

Results

The primary olfactory organ in *Drosophila* is the antenna, which houses hundreds olfactory sensilla on the surface of its third segment (**Figure 1—figure supplement 1A** [↗](#)). Each sensillum typically encapsulates the outer dendrites of two to four ORNs. The outer dendrites are the sites where odorant receptors are expressed, enabling the detection of volatile chemicals. A small portion of the outer dendrites lies beneath the base of the sensillum cuticle. At the ciliary constriction, the outer dendrites connect to the inner dendritic segment, which then links to the soma of each ORN (**Figure 1—figure supplement 1B** [↗](#)) (Nava Gonzales et al., 2021 [↗](#); Ng et al., 2020 [↗](#); Shanbhag et al., 1999 [↗](#); Su et al., 2009 [↗](#)).

To survey *Drosophila* CO₂-sensing ab1C neurons at the population level, we first generated an SBEM volume covering the antennal region where ab1 sensilla are located, named the ab1 zone, on the proximal medial surface of the third antennal segment (**Figure 1A** [↗](#), and **Figure 1—figure supplement 1A** [↗](#)) (de Bruyne et al., 2001 [↗](#); Grabe et al., 2016 [↗](#); Shanbhag et al., 1999 [↗](#)). Notably, ab1 is the only antennal large basiconic sensillum type that houses four ORNs (Shanbhag et al., 1999 [↗](#)), allowing us to unequivocally identify individual ab1 sensilla based on their relatively large size and distinctive set of four associated neurons without having to rely on genetic labeling.

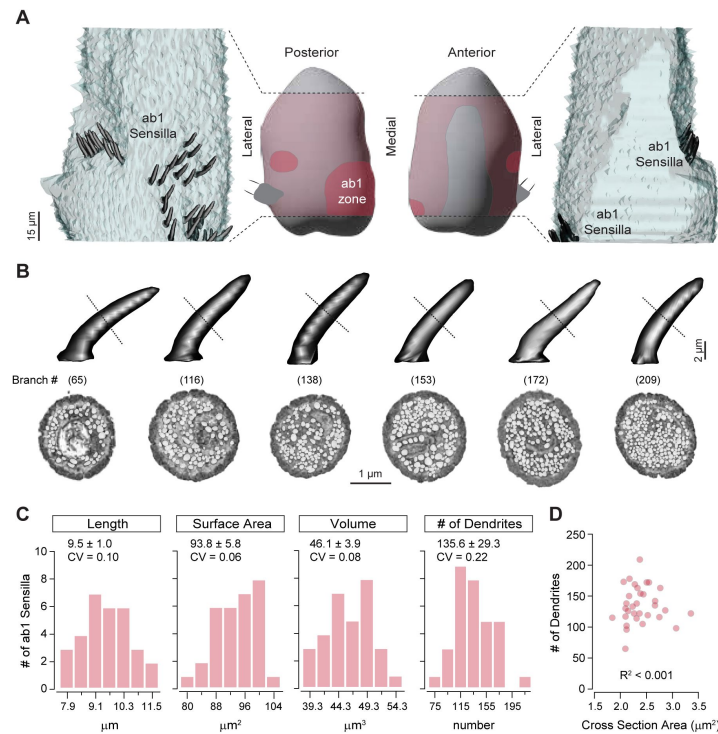


Figure 1.

Distribution of ab1 sensilla and their morphometric analysis

(A) Middle panels: Illustration of the antennal region sampled in the SBEM volume (pink) with the ab1 zone highlighted in crimson. Left and right panels: 3D models of ab1 sensillum cuticles (dark gray) are shown on the imaged portion of the antenna (light gray). Scale bar: 15 μm .

(B) 3D models and corresponding SBEM images of ab1 sensilla. Dashed lines indicate the approximate mid-point region of cuticles where the SBEM images were sampled. Sensilla are arranged from left to right in order of increasing dendritic branch counts, as indicated in parentheses. Dendrites are pseudocolored in white. Scale bars: 2 μm for 3D models and 1 μm for SBEM images.

(C) Distribution of morphometric features (length, surface area, volume, and dendritic branch counts) from fully segmented ab1 sensillum cuticles. Mean $\pm$ SD and coefficients of variation (CV) are shown above each graph ($n = 31\text{--}32$).

(D) Correlation analysis of ab1 dendritic branch counts as a function of sensillum mid-point cross-sectional areas ($n = 32$). See Table S1 for source data.

Our survey of the antennal SBEM volume identified 39 ab1 sensilla, accounting for over 80% of the total ab1 population (48 sensilla), as previously documented (Shanbhag et al., 1999). Instead of being uniformly distributed within the ab1 zone, the ab1 sensilla were arranged in two clusters: a larger, more diffuse cluster on the proximal medial surface and a smaller cluster on the lateral side of the antennal surface (Figure 1A).

Interestingly, we observed substantial variation in the number of dendritic branches in the ab1 sensilla, ranging from 65 to 295 branches across 32 sensilla with quality EM images that allowed for clear distinction of individual dendrites (Figure 1B,C). These numbers were estimated by counting dendrites around the mid-point of the sensilla, yielding an average of 136 ± 29 branches (mean $\pm$ SD, $n=32$, representing 67% of the ab1 population). This result aligns with a previous transmission electron microscopy (TEM) study, which reported an average of 95 ± 12 ab1 dendritic branches from images at unspecified positions along the longitudinal sensillum axis (Shanbhag et al., 1999). The wide range of dendritic branch counts, reflected by a coefficient of variation (CV) of 0.22, indicates considerable heterogeneity among ab1 sensilla.

Does the heterogeneity in dendritic counts reflect variability in ab1 sensillum cuticle morphology? To address this question, we measured the length (L), surface area (SA), and volume (V) of 31 intact ab1 sensilla, excluding those with truncated structures. The ab1 cuticle morphometric features ($L = 9.50 \pm 0.98 \mu\text{m}$; $SA = 93.75 \pm 5.83 \mu\text{m}^2$; $V = 46.09 \pm 3.91 \mu\text{m}^3$, $n=31$) were comparable to our published ab1 measurements from two independent SBEM volumes ($L = 10.80 \pm 0.16 \mu\text{m}$; $SA = 70.70 \pm 0.55 \mu\text{m}^2$; $V = 39.49 \pm 2.12 \mu\text{m}^3$, $n=5$) (Nava Gonzales et al., 2021). Although these morphometric features also exhibited variation across the ab1 population, their coefficients of variation were markedly smaller ($CV_L = 0.10$, $CV_{SA} = 0.06$, $CV_V = 0.08$) compared to that of dendritic branch counts ($CV_{DC} = 0.22$) (Figure 1C). This analysis suggests that sensillum cuticle morphology is less variable than dendritic counts. Furthermore, sensilla with higher dendritic counts are not necessarily thicker than those with fewer branches (see Table S1 for source data). Indeed, an analysis of mid-point dendritic number versus sensillum cuticle cross-sectional area revealed no correlation ($R^2 \approx 0$, Figure 1D), suggesting that cuticular morphology and ORN dendritic number are regulated independently.

Olfactory receptor neurons in the ab1 sensillum

The ab1 sensillum houses four ORNs (Figure 2A), each expressing a specific receptor or receptor complex: Or42b (ab1A), Or92a (ab1B), Gr21a/Gr63a (ab1C), and Or10 (ab1D) (Couto et al., 2005; de Bruyne et al., 2001; Hallem et al., 2004; Jones et al., 2007; Kwon et al., 2007). These neurons exhibit distinct extracellular spike amplitudes (Figure 2A), with a relative size ratio of 5.2 (ab1A): 4.5 (ab1B): 2.3 (ab1C): 1 (ab1D) (Zhang et al., 2019). Our published studies have shown that, within a sensillum, the rank order of ORN sizes reflects their relative extracellular spike amplitudes (Nava Gonzales et al., 2021; Tsang et al., 2018; Zhang et al., 2019). This allowed us to assign the cellular identities of ab1 ORNs based on their relative neuronal sizes without having to rely on genetic labeling.

We successfully segmented all four ORNs from an ab1 sensillum (Figure 2A). As expected, the co-housed ORNs differed in size, and based on their sizes in descending order, we assigned neuronal identities as ab1A, B, C, and D (Figure 2B). Similar to the large-spike neurons in the other two large basiconic sensilla (ab2A and ab3A), both ab1A and ab1B displayed numerous dendritic branches, with total counts of 67 and 37, respectively. Given that the segmented ab1 sensillum had the fewest total dendritic branches, it is likely that other ab1A and ab1B neurons exhibit more numerous dendritic branches. As with other basiconic ORNs, the branching patterns of ab1A and ab1B were complex, featuring multiple branching points. From the base of the sensillum cuticle to its tip, the primary branches of ab1A or ab1B neurons bifurcated multiple times, generating secondary branches that further divide into tertiary, quaternary, or higher-order

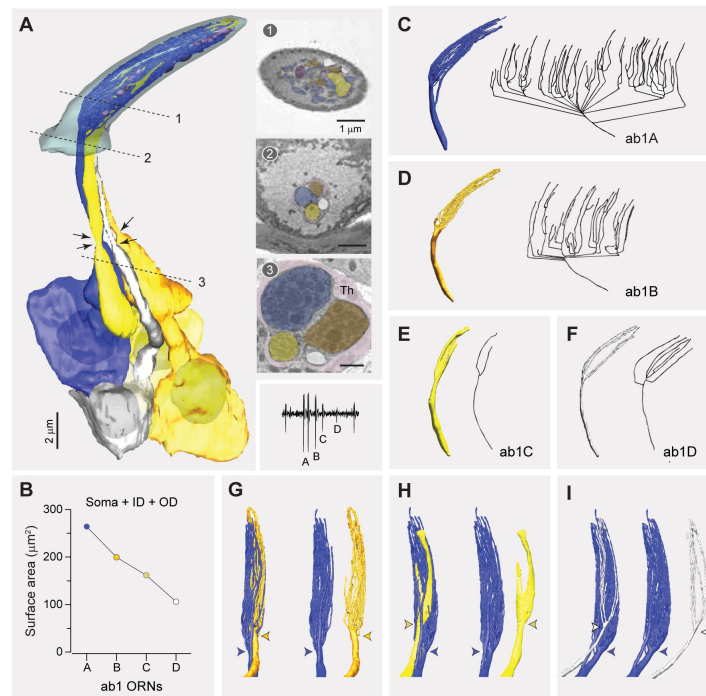


Figure 2.

Fully reconstructed ab1 sensillum with four neurons

(A) 3D model and SBEM images of a fully reconstructed ab1 sensillum. ORNs are pseudocolored to indicate neuronal identities: ab1A (blue), ab1B (orange), ab1C (yellow) and ab1D (white). Dashed lines mark the positions of corresponding SBEM images on the right: (1) sensillum lumen containing the outer dendrites; (2) proximal region of the outer dendrites; and (3) the inner dendrites surrounded by processes of the thecogen cell (Th). Arrows indicate the positions of ciliary constriction, which demarcates the inner and outer dendritic segments. Inset: a representative trace from single-sensillum recording showing the relative extracellular spike amplitudes of the ab1 ORNs. Scale bars: 2 μm for 3D models and 1 μm for SBEM images.

(B) Combined surface areas of the ORN soma, inner dendrite (ID), and outer dendrites (OD).

(C–F) 3D models and corresponding 2D projections of the outer dendritic branches of ab1A (C), ab1B (D), ab1C (E), and ab1D (F).

(G–I) Spatial relationship between the dendritic branches ab1A and those of its neighboring neurons. Colored arrow heads indicate the primary branching points or flattening position.

See also Video S1.

branches (**Figure 2C,D**). This complex branching pattern explains why the mid-point dendritic count of this sensillum (65 branches) was lower than the combined total of 104 branches for its ab1A and ab1B.

In contrast, the neighboring ab1C and ab1D neurons exhibited simpler branching patterns, characterized by fewer branches and a lower degree of branching (ab1C: 2 and ab1D: 7 branches; **Figure 2E,F**). Notably, the outer dendrite of the CO₂-sensing ab1C had a flattened, sheet-like morphology in its distal region (**Figure 2E**), contrasting with the narrow, cylindrical branches of its odor-sensing neighbors (**Figure 2C,D,F**). Interestingly, this dendritic sheet appeared to split into two (**Figure 2E**), distinct from the intact ab1C dendritic sheet reconstructed from another SBEM volume (**Figure 2—figure supplement 1**). This observation suggests the possibility of morphological diversity among ab1C dendrites, a question we addressed in detail in the following section.

The outer (sensory) dendrites, starting at the ciliary constriction, comprised two distinct segments: an unbranched cylindrical segment located below the sensillum cuticle base and a branched (or flattened) segment situated within the sensillum lumen (**Figure 2A**). Of note, the outer dendrite of ab1A began branching more proximally near the cuticle base compared to the neighboring neurons (**Figure 2G-I**), unlike other types of co-housed basiconic ORNs, whose dendrites typically start branching at a similar level near the sensillum cuticle base (Nava Gonzales et al., 2021). Furthermore, the dendritic branches of ab1A appeared to envelop those of the neighboring neurons, occupying the circumference region of the sensillum lumen (Video 1). We observed similar dendritic branching features in another ab1 sensillum partially sampled from a previously generated antennal SBEM volume (**Figure 2—figure supplement 1**), suggesting that the described morphologies are representative of ab1 ORNs.

Morphological diversity across ab1C outer dendrites

To investigate the potential diversity in ab1C dendritic morphologies, we segmented the CO₂-sensing neurons from all ab1 sensilla with high-quality images for 3D reconstruction. Indeed, our results revealed remarkable morphological heterogeneity among individual ab1C neurons. Specifically, dendritic flattening occurred at positions ranging from 13% to 53% of the cuticle length above the sensillum base, with an average position of 40%, while dendritic termination varied between 62% to 98% of the cuticle length, averaging 89%. In general, dendrites that began flattening closer to the base tended to terminate earlier before reaching the cuticle tip (**Figure 3A**). Furthermore, these diverse dendritic morphologies can be categorized into four types (**Figure 3A**, inset) as detailed below (also see Video 2).

Loosely curled category

The majority of ab1C dendritic sheets fell into the “loosely curled” category, where the longitudinal edges of the flattened segment gently curved inward in a single intact structure. In EM cross-sections, these dendritic sheets typically exhibited a “C”-shaped profile (Images 1 & 2, **Figure 3B**). Some cross-sections resembled an inverted “U”, as if the dendritic sheet had folded along the longitudinal axis (Image 3); or an “S” profile when the two edges curled in opposite directions (Image 4). In one instance, a central ridge extended along the sheet’s longitudinal axis to form a raised protrusion on the otherwise flat dendritic structure (Image 5, **Figure 3B**).

Fully curled category

Surprisingly, a sizeable minority of ab1C dendritic sheets formed completely curled, ring-shaped or tube-like structures that encircled several dendritic branches from neighboring neurons, forming a “dendrite-within-dendrite” structure (**Figure 3C**). Within this category, the cross-section profiles of dendritic sheets varied from a completely enclosed ring (Image 1) to a lasso-like shape (Image 2), and even a more intricate “θ”-like configuration (Image 3, **Figure 3C**).

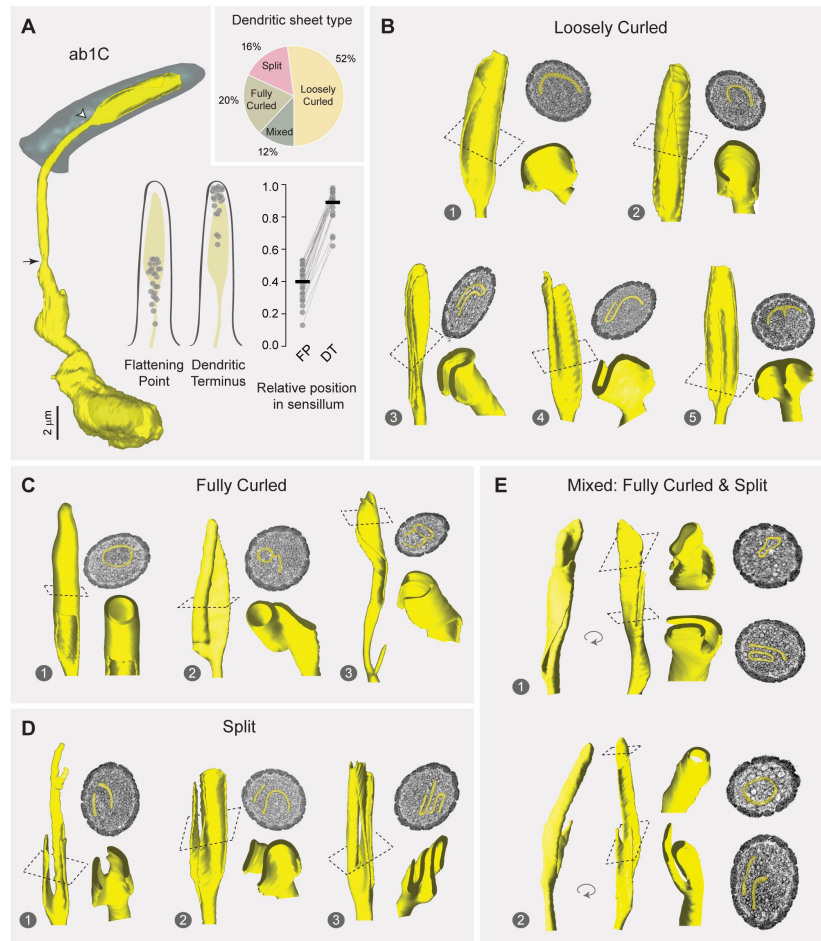


Figure 3.

Morphological diversity across ab1C outer dendrites

(A) Left panel: 3D model of an ab1C neuron (yellow) and its associated sensillum cuticle (gray). Arrow indicates ciliary constriction, marking the beginning of the outer (sensory) dendrite. White arrow head marks the location where dendritic flattening occurs. Middle panel: Positions of the flattening point (FP) and dendritic terminus (DT) relative to the cuticle length. Filled gray circles represent the relative positions of individual ab1C neurons. Right panel: Similar to the middle panel, but with lines connecting data points from the same neurons. The cuticle base and tip are designated as 0 and 1, respectively. Horizontal bars indicate the mean positions ($n = 25$). Inset: Pie chart illustrating the distribution of the four outer dendritic morphological categories across the ab1C population.

(B–E) Representative 3D models, top-down clipped views, and corresponding cross-sectional EM images are shown for each of the four morphological categories: loosely curled **(B)**, fully curled **(C)**, split **(D)**, and mixed **(E)**. ab1C dendritic sheets are pseudocolored in yellow in sample EM images.

See also Video S3.

Interestingly, similar ring-shaped or complex dendritic profiles were previously observed in TEM images of a large basiconic sensillum containing four neurons, although the neuronal identity was undetermined (Shanbhag et al., 1999 [↗](#)).

Split category

Another subset of ab1C dendritic sheets split into two or more distinct parts (**Figure 3D** [↗](#), also see **Figure 2E** [↗](#)). Within this category, the dendrites also exhibited remarkable morphological heterogeneity. For example, one dendrite split longitudinally into two parts, with one further bifurcating at the distal region (Image 1). Another example displayed an uneven division, with a smaller rectangular sheetlet and a larger one resembling a loosely curled structure (Image 2). A third example demonstrated a complex morphology, with both sheetlets curling into a “U” shape in opposite directions (Image 3, **Figure 3D** [↗](#)).

Mixed category

The fourth morphological type exhibited the most intricate structure and was classified as the “mixed” category. These dendritic sheets combined features of both the “fully curled” and “split” categories (**Figure 3E** [↗](#)). In one instance, the dendritic sheet divided into two parts: one formed a fully enclosed tubular structure, while the other sheetlet wrapped around this tube (Image 1). In another example, one dendritic sheetlet formed a fully enclosed, blind-ended tube at the distal end, whereas the other terminated prematurely with a slight bifurcation, extending only about one-third the length of the first sheetlet (Image 2, **Figure 3E** [↗](#)).

ab1C morphometrics

How might dendritic flattening influence sensory function? To address this question, we compared the morphometric features of the cylindrical and flattened segments of ab1C outer dendrites (**Figure 4A** [↗](#), left panel, Segments 1 and 2, respectively). The cylindrical segment extended from the ciliary constriction below the sensillum cuticle base to approximately 40% of the cuticle length above the base, where dendritic flattening occurred (**Figures 3A** [↗](#)). On average, the flattened segment was half as long as the cylindrical segment, with average lengths of 5.74 μm and 10.72 μm , respectively. Despite its shorter length, the surface area of the flattened segment was about 22% larger than the cylindrical counterpart. In contrast, the volume of the flattened segment was much smaller (1.03 μm^3), only about 40% of the cylindrical counterpart (2.64 μm^3 , **Figure 4A** [↗](#) and **Table 1** [↗](#)).

These morphometric comparisons suggest that dendritic flattening increases the surface area-to-volume ratio. This could potentially enhance the dendrite’s capacity to detect CO_2 and facilitate signal propagation as the increased surface area allows for more efficient passive spread of electrical signals along the dendrite. Indeed, the average volume per unit length (or cross-sectional area) of the flattened segment was about 28% smaller than that of the cylindrical segment (0.18 vs. 0.25 μm^2 , **Figure 4B** [↗](#), and **Table 1** [↗](#)), indicating modest tapering of the dendrite towards its tip. Strikingly, the surface area-to-volume ratio of the flattened segment was over 300% as high as the cylindrical segment (23.94 vs. 7.58, **Figure 4C** [↗](#), and **Table 1** [↗](#)). Notably, none of the morphometric features differ significantly among ab1C dendrites in different morphological categories (**Figure 4D** [↗](#)), implying that the morphological diversity may have evolved to serve additional functions. Overall, the marked increase in the surface area-to-volume ratio underscores the potential functional advantages of dendritic flattening in sensory processes.

Heterogeneity in ab1D branching patterns

For comparison, we analyzed the neighboring odor-sensing ab1D, selected for its relatively low number of dendritic branches, making it suitable for systematic analysis. As previously noted, ab1D was identified based on its smallest soma and inner dendrite among the four ab1 ORNs (see

Figure 4.

Flattening increases the surface area-to-volume ratio of the ab1C outer dendrite

(A) Left panel: Illustration of an ab1C neuron. Arrow indicates the position of ciliary constriction. White arrowhead marks the location where dendritic flattening occurs. Segment #1: cylindrical outer dendrite; Segment #2: flattened outer dendrite. Right panels: Distributions of length, surface area and volume for the cylindrical (coral) or flattened (teal) outer dendritic segments across the ab1C population (n = 25). (B–C) Quantitative comparisons of the volume per unit length (B) and surface area-to-volume ratio (C) for the cylindrical (coral) and flattened (teal) outer dendritic segments. Each filled circle represents data from an individual neuron, with horizontal bars indicating the mean values (n = 25). Statistical significance was determined using paired two-tailed *t*-test.

(D) Quantitative comparisons of morphometric features for ab1C flattened segments across different morphological categories. Each filled circle represents data from an individual ab1C neuron, with horizontal bars indicating the mean values. Statistical significance is determined by Kruskal-Wallis one-way ANOVA on ranks and denoted by different letters. For example, labels “a” and “b” indicate a significant difference between groups ($P < 0.05$), whereas labels with identical or shared letters (e.g., “a” and “a”, “a,b” and “a”, or “a,b” and “b”) indicate no significant difference. See Table S1 for source data.

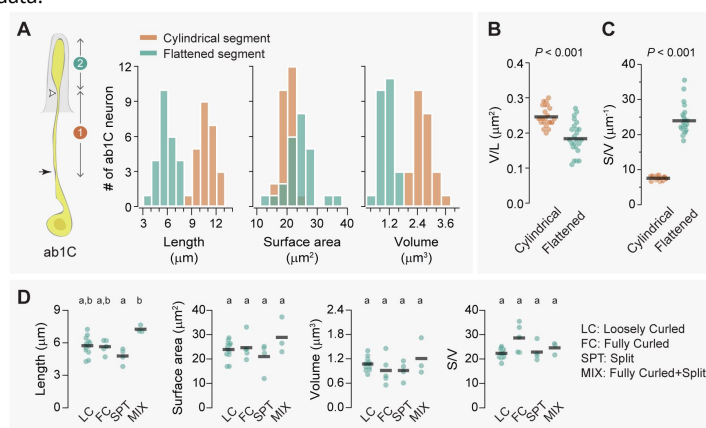


Table 1.

Morphometric data for the outer dendritic segments of ab1C and ab1D (mean $\pm$ SD)

ORN type	Length (μm)			Surface area (μm^2)			Volume (μm^3)			V/L			S/V		
ab1C	Proximal	Distal	Total	Proximal	Distal	Total	Proximal	Distal	Total	Proximal	Distal	Total	Proximal	Distal	Total
<i>Loosely Curled</i> (n=13)	10.67 $\pm$ 0.86	5.73 $\pm$ 0.84	16.40 $\pm$ 1.41	19.82 $\pm$ 2.10	23.90 $\pm$ 3.69	43.72 $\pm$ 3.91	2.60 $\pm$ 0.34	1.07 $\pm$ 0.16	3.68 $\pm$ 0.36	0.24 $\pm$ 0.02	0.19 $\pm$ 0.04	0.22 $\pm$ 0.02	7.64 $\pm$ 0.44	22.32 $\pm$ 1.96	11.93 $\pm$ 0.93
<i>Fully Curled</i> (n=5)	11.62 $\pm$ 0.88	5.64 $\pm$ 0.62	17.27 $\pm$ 0.78	20.86 $\pm$ 2.03	24.66 $\pm$ 5.03	45.52 $\pm$ 4.37	2.73 $\pm$ 0.26	0.91 $\pm$ 0.35	3.64 $\pm$ 0.43	0.24 $\pm$ 0.03	0.16 $\pm$ 0.06	0.21 $\pm$ 0.03	7.67 $\pm$ 0.72	28.61 $\pm$ 5.80	12.56 $\pm$ 1.07
<i>Split</i> (n=4)	10.76 $\pm$ 0.59	4.78 $\pm$ 0.71	15.54 $\pm$ 0.48	20.33 $\pm$ 2.61	21.00 $\pm$ 6.12	41.33 $\pm$ 7.32	2.80 $\pm$ 0.49	0.91 $\pm$ 0.23	3.71 $\pm$ 0.66	0.26 $\pm$ 0.03	0.20 $\pm$ 0.07	0.24 $\pm$ 0.05	7.32 $\pm$ 0.48	22.88 $\pm$ 3.82	11.21 $\pm$ 1.60
<i>Mixed</i> (n=3)	9.38 $\pm$ 0.68	7.26 $\pm$ 0.33	16.64 $\pm$ 0.93	18.07 $\pm$ 1.43	28.92 $\pm$ 7.44	47.00 $\pm$ 6.73	2.40 $\pm$ 0.16	1.21 $\pm$ 0.45	3.61 $\pm$ 0.30	0.26 $\pm$ 0.03	0.17 $\pm$ 0.07	0.19 $\pm$ 0.02	7.54 $\pm$ 0.66	24.58 $\pm$ 2.51	12.99 $\pm$ 0.84
Total (n=25)	10.72 $\pm$ 0.97	5.74 $\pm$ 0.95	16.47 $\pm$ 1.19	19.90 $\pm$ 2.09	24.19 $\pm$ 4.89	44.09 $\pm$ 4.81	2.64 $\pm$ 0.33	1.03 $\pm$ 0.25	3.67 $\pm$ 0.39	0.25 $\pm$ 0.03	0.18 $\pm$ 0.05	0.22 $\pm$ 0.03	7.58 $\pm$ 0.50	23.94 $\pm$ 3.93	12.07 $\pm$ 1.11
ab1D															
<i>Unbranched</i> (n=8)	NA	NA	17.47 $\pm$ 1.85	NA	NA	13.29 $\pm$ 1.18	NA	NA	0.69 $\pm$ 0.08	NA	NA	0.040 $\pm$ 0.006	NA	NA	19.50 $\pm$ 2.13
<i>Branched</i> (n=13)	8.94 $\pm$ 1.11	29.09 $\pm$ 10.25	38.83 $\pm$ 10.05	8.97 $\pm$ 1.72	10.77 $\pm$ 3.30	19.74 $\pm$ 4.12	0.60 $\pm$ 0.10	0.30 $\pm$ 0.10	0.90 $\pm$ 0.20	0.068 $\pm$ 0.014	0.010 $\pm$ 0.003	0.025 $\pm$ 0.006	15.03 $\pm$ 1.63	37.51 $\pm$ 4.18	22.25 $\pm$ 2.65
Total (n=21)			30.20 $\pm$ 12.78			17.28 $\pm$ 4.57			0.82 $\pm$ 0.18			0.030 $\pm$ 0.010			21.20 $\pm$ 2.81

Figure 2B [for example](#)). To investigate potential heterogeneity in ab1D dendritic morphologies, we focused on ab1 sensilla with high-quality images for further analysis.

We successfully generated 3D models for 21 ab1D neurons, and found that their average soma volume was markedly smaller than that of ab1C ($23.73 \mu\text{m}^3$ vs. $42.68 \mu\text{m}^3$). These models revealed significant heterogeneity in dendritic branching patterns, from unbranched to branched outer dendrites (**Figure 5A** [inset](#)). Among the 13 branched ab1D neurons, the primary branching point was located at various positions ranging from 8% to 44% above the sensillum base relative to the cuticular length (mean $\pm$ SD = $21 \pm 10\%$, CV = 0.46). Dendritic termination, on the other hand, was less variable (CV = 0.09) and occurred between 63% to 97% of the cuticle length, with an average of $91 \pm 8\%$ (**Figure 5A** [inset](#)). Dendrites that began branching closer to the cuticle base did not necessarily terminate earlier (**Figure 5A** [inset](#)), differing from the relationship observed between ab1C flattening and dendritic termination locations (**Figure 3A** [inset](#)).

Interestingly, eight ab1D neurons had a simple, unbranched outer dendrite (**Figure 5B** [inset](#)), unlike most characterized basiconic ORNs, which typically have multiple dendritic branches (Nava Gonzales et al., 2021 [inset](#)). In contrast, the remaining 13 ab1D neurons exhibited sparsely branched dendrites (**Figure 5C** [inset](#)). Notably, these neurons exhibited heterogeneity in both branching patterns and the number of branches, which ranged from two to seven (**Figure 5D** [inset](#)). This variability in dendritic branching patterns mirrors the morphological heterogeneity observed in ab1C dendrites, suggesting that both types of neurons may have evolved diverse structural features to support specialized sensory functions.

ab1D morphometrics

Unbranched vs. branched ab1D neurons

We then compared the morphometrics of unbranched outer dendrites with those of branched ones (**Figure 6A–E** [inset](#), comparisons between segment 1). The summed measurements encompass the entire outer dendritic region where olfactory receptors are localized. The ab1D neurons with unbranched dendrites have an average outer dendritic length of $17.47 \mu\text{m}$, sensory surface area of $13.29 \mu\text{m}^2$, and volume of $0.69 \mu\text{m}^3$. In comparison, ab1D with branched dendrites exhibited greater summed outer dendritic length ($38.03 \mu\text{m}$), surface area ($19.74 \mu\text{m}^2$), and volume ($0.89 \mu\text{m}^3$, **Table 1** [inset](#)). Notably, the dendritic surface area-to-volume ratio of neurons with branched dendrites was only about 14% higher than that of unbranched neurons (22.25 vs. 19.50, **Table 1** [inset](#)). These comparisons suggest that within the ab1D neuronal population, branching markedly increases the total sensory surface area as much as 49% ($19.74 \mu\text{m}^2$ vs. $13.29 \mu\text{m}^2$), while providing only a modest enhancement in the overall surface area-to-volume ratio (**Table 1** [inset](#)).

Branched ab1D neurons: proximal vs. distal outer dendrites

For the branched ab1D neurons, we further compared the morphometrics of their proximal (unbranched) and distal (branched) outer dendritic segments (**Figure 6A–E** [inset](#), comparisons between segments 2 & 3). Within a single neuron, the branched distal segment exhibited an average total length over 200% longer than the proximal unbranched segment, while showing a modest 20% increase in total surface area (**Figure 6A–B** [inset](#)). In addition, the distal segments had a significantly smaller total volume (distal/proximal $\approx 50\%$, **Figure 6C** [inset](#)), resulting in a much smaller average volume per unit length or cross-sectional area (distal/proximal $\approx 15\%$, **Figure 6D** [inset](#)). This indicates substantial narrowing of the branched outer dendrite, with the distal segment's average diameter being approximately 40% of the proximal segment's diameter. Consequently, the surface area-to-volume ratio of the distal branched segments was significantly higher—about 250% that of the proximal unbranched segments (**Figure 6E** [inset](#) and **Table 1** [inset](#)). In all, our analysis suggests that when comparing the distal and proximal dendritic segments for the

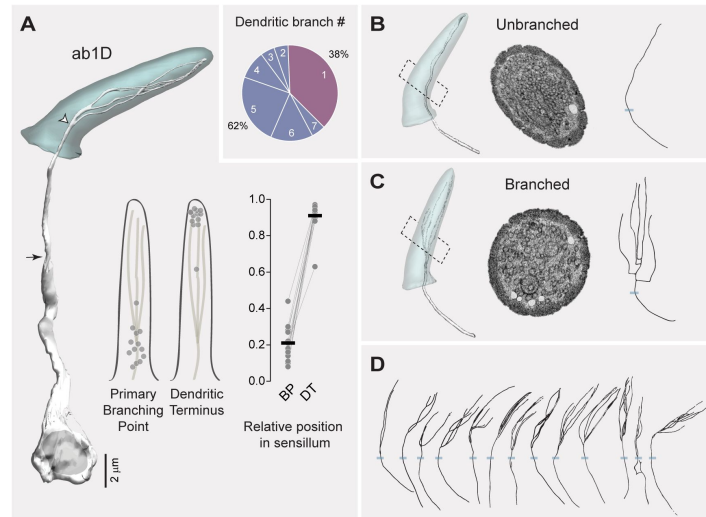


Figure 5.

Heterogeneity in ab1D branching pattern

(A) Left panel: 3D model of an ab1D neuron (white) and its associated sensillum cuticle (gray). Arrow indicates ciliary constriction, marking the beginning of the outer (sensory) dendrite. White arrow head marks the primary branching point where dendritic branching starts. Middle panel: Positions of the primary branching point (BP) and dendritic terminus (DT) relative to the cuticle length. Filled gray circles represent the relative positions of individual ab1D neurons with branched dendrites. Right panel: Similar to the middle panel, but with lines connecting data points from the same neurons. The cuticle base and tip are designated as 0 and 1, respectively. Horizontal bars indicate the average positions of the segmented ab1D neurons ($n = 13$). Inset: Pie chart illustrating the distribution of the two dendritic branching categories across the ab1D population ($n = 21$).

(B–C) Representative 3D models, corresponding cross-sectional EM images, and 2D projections of dendritic skeletons are shown for the two morphological categories: unbranched **(B)**, and branched **(C)**. Blue bars mark the positions of cuticular bases. ab1D dendrites are pseudocolored in white in sample EM images.

(D) Dendritic skeletons for the 13 segmented ab1D neurons with branched outer dendrites, with increasing branch numbers arranged from left to right.

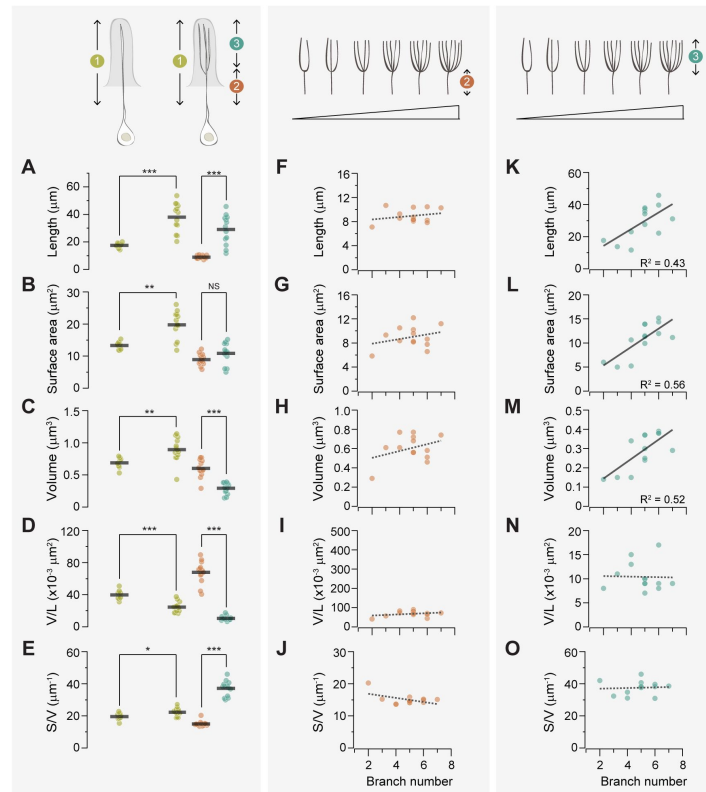


Figure 6.

Branching enhances the surface area-to-volume ratio of the ab1D outer dendrite

(A-E) Quantitative comparisons of outer dendrites morphometric properties, including total length (A), surface area (B), volume (C), volume per unit length (D), and surface area-to-volume ratio (E). Data are presented for the entire outer dendrite (summed morphometrics, segment 1 in mustard), proximal unbranched (segment 2 in coral), and distal branched segments (segment 3 in teal). For branched ab1D neurons, segment 3 represents the summed morphometric measurements of all dendritic branches above the primary branch point, while segment 1 includes the combined values of segments 2 and 3. Each filled circle represents data from an individual neuron, with horizontal bars indicating the mean values ($n = 8$ for unbranched neurons; $n = 13$ for branched neurons). Statistical significance was determined using Mann-Whitney rank sum test for comparisons between segment 1, and paired t -test for comparison between segments 2 and 3. * $P < 0.05$; ** $P < 0.01$; *** $P < 0.005$.

(F-J) For branched ab1D dendrites, correlation analysis of the morphometric properties of the proximal outer dendrite (segment 2, coral) in relation to the number of dendritic branches. Parameters include the length (F), surface area (G), volume (H), volume per unit length (I), and surface area-to-volume ratio (J). (K-O) Similar to (F-J) but with the summed morphometric measurements of the distal outer dendritic segments (segment 3, teal). Linear fits are shown, with dashed lines indicating $R^2 < 0.4$ and solid lines indicating $R^2 > 0.4$. See Table S1 for source data.

same ab1D neuron, sparse branching (≤ 7 branches) has a modest impact on the overall sensory surface area but substantially increases the surface area-to-volume ratio in the outer dendritic region.

Impacts of branch number on proximal dendritic morphometrics

Our morphometric analysis of 13 sparsely branched ab1D neurons allowed us to determine how branch number influences the morphometric properties of outer dendrites in an ORN population. First, we examined whether the distal dendritic branch number affects the morphometrics of the proximal unbranched segment, and found no significant impact (**Figure 6F–J**). These findings indicate that when the branch number is low, the morphometrics of the proximal “trunk” region remain relatively consistent.

Impacts of branch number on distal dendritic morphometrics

In contrast to the proximal dendritic morphometrics, we found that the total length, surface area, and volume of distal segments scaled with the number of dendritic branches (**Figure 6K–M**). Interestingly, the diameter of individual dendritic branches and the surface area-to-volume ratio remained relatively consistent (**Figure 6N–O**), suggesting that the dimensions of individual dendritic branch are conserved regardless of the branch number. The average dimensions of an ab1D distal dendritic branch are $6.00\ \mu\text{m}$ in length, $2.22\ \mu\text{m}^2$ in surface area, and $0.06\ \mu\text{m}^3$ in volume. Therefore, increasing the number of dendritic branches expands the total sensory surface area without altering the surface area-to-volume ratio, thereby preserving the dendrite’s electrical signal propagation properties (see Discussion). Our analysis suggests that structural scaling in ab1D neurons may enhance sensory capacity while preserving the biophysical properties of dendrites.

Auxiliary cells in the ab1 sensillum

In addition to ORNs, each olfactory sensillum contains three types of auxiliary cells: thecogen, trichogen, and tormogen cells. In previous work, we characterized the 3D morphology of these cells in the ab4 and ac3 sensilla (Nava Gonzales et al., 2021). To determine whether auxiliary cells exhibit sensillum-type-specific morphologies, we extended our analysis to the ab1 sensillum (**Figure 7A** and Video S3).

Similar to the ab4 and ac3 sensilla, the three auxiliary cells in the ab1 sensillum followed a conserved cellular organization. The thecogen cell exhibited an overall flat morphology, appearing to adhere closely to the ORNs. The cell formed a tight sleeve around the entire outer dendritic region beneath the cuticle base, as well as the inner dendrites, and portions of the ORN somas. Notably, in the inner dendritic region, the thecogen cell separated individual ORNs from one another (**Figure 7A–C**). This differs from the ab4 sensillum, where the thecogen cell encloses both inner dendrites within a single bundle without separating them (Nava Gonzales et al., 2021).

The ab1 trichogen cell was positioned distal and lateral to the thecogen cell, with cellular processes surrounding the distal region of thecogen cell. The trichogen cell also featured an apical surface with extensive microlamellae bordering the sensillum-lymph cavity (**Figure 7A**, B, and D). The ab1 tormogen cell, the outermost of the three, partially enveloped the trichogen cell near the cuticle base. Its nucleus was located at the level of the ORN inner dendrites. Interesting, unlike its counterparts in the ab4 and ac3 sensilla (Nava Gonzales et al., 2021), this tormogen cell lacked a stalk-like protrusion from its basal region, suggesting that this structure is not a universal feature of tormogen cells in insect olfactory sensilla.

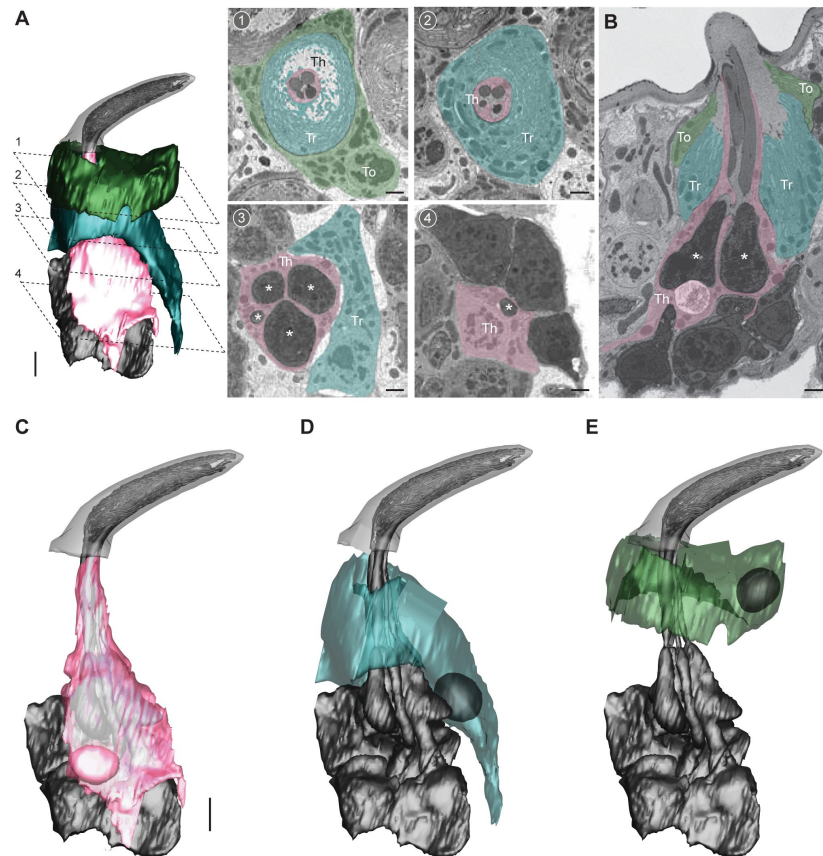


Figure 7.

Auxiliary cells in the *ab1* sensillum

(A) 3D model and SBEM images of an *ab1* sensillum. Cells are pseudocolored to indicate identities: ORNs (gray), thecogen cell (pink, Th), trichogen cell (turquoise, Tr), tormogen cell (green, To). Dashed lines mark positions of the corresponding SBEM images: (1–2) outer dendritic region beneath the cuticle; (3) inner dendrites (asterisks); (4) ORN somas.

(B) A longitudinal sensillum section rendered by IMOD.

(C–E) 3D models of individual auxiliary cells: thecogen (C), trichogen (D), and tormogen (E). For simplicity, microlamellae of trichogen and tormogen cells were not segmented and thus not shown in 3D models.

Scale bars: 2 μ m for 3D models and 1 μ m for SBEM images. See also Video S3.

Discussion

Our nanoscale, population-wide analysis of the outer dendritic structures of ab1C and ab1D neurons revealed significant morphological diversity between different neuronal types and among homotypic ORNs. Between heterotypic ORNs, we identified a striking structural distinction between the CO₂-sensing ab1C neurons and the odor-sensing ab1D neurons. Specifically, ab1C neurons displayed flattened, sheet-like dendrites, in contrast to the cylindrical branches observed in ab1D neurons. Notably most *Drosophila* ORNs, whether detecting food odors or fly pheromones, have outer dendrites with a cylindrical morphology (Nava Gonzales et al., 2021 [DOI](#)). Therefore, the sheet-like outer dendrites of ab1C neurons likely represent a unique structural adaptation for CO₂ or gas sensing in insects. Indeed, TEM studies in mosquitoes, moths, sand flies, and beetles have documented flattened dendritic lamellae, which are likely associated with CO₂-sensing neurons (Hull & Cribb, 1997 [DOI](#); Lu et al., 2007 [DOI](#); S. B. McIver, 1972 [DOI](#); S. McIver & Siemicki, 1975 [DOI](#); Stange et al., 1995 [DOI](#); Stange & Stowe, 1999 [DOI](#); White et al., 1974 [DOI](#)).

What might be the functional significance of ab1C's unique dendritic architectures? While both dendritic flattening and branching can expand sensory surface area and enhance SA/V ratio (Figures 4 [DOI](#) and 6 [DOI](#)), the flattened ab1C dendritic sheets may facilitate better distribution or effective surface expression of the Gr21a and Gr63a receptors (Jones et al., 2007 [DOI](#); Kwon et al., 2007 [DOI](#); Suh et al., 2004 [DOI](#)). Notably, ectopic expression of Gr21a and Gr63a in the ab3A “empty” neurons can confer CO₂ responsiveness, but the sensitivity is orders of magnitude lower than that of endogenous ab1C neurons (Kwon et al., 2007 [DOI](#); Yao & Carlson, 2010 [DOI](#)), despite ab3A neurons having a much larger sensory surface area (ab3A: 145 μm²; ab1C: 44 μm²) (Nava Gonzales et al., 2021 [DOI](#) and this study). This observation suggests that the reduced CO₂ sensitivity in ab3A neurons may result from insufficient surface expression of Gr21a and Gr63a.

Consistent with this, CO₂ sensitivity can be markedly enhanced by increasing the copy number of either Gr21a or Gr63a transgene, although not to the level observed in endogenous ab1C neurons (Kwon et al., 2007 [DOI](#)). In future research, it will be interesting to determine the molecular mechanisms underlying ab1C's flattened dendritic morphology and explore how selective manipulation of this feature affects CO₂ detection.

Beyond the morphological diversity observed between heterotypic ORNs, our study uncovered a rich spectrum of dendritic architectures within homotypic ab1C or ab1D neuron populations. For example, ab1C outer dendrites exhibit a wide range of morphologies, including plain sheets, tube-like structures enclosing several neighboring dendrites, split dendritic sheetlets, and combinations of all of these features (Figure 3 [DOI](#)). Similarly, ab1D dendrites range from simple, unbranched forms to numerous branched morphologies (Figure 5 [DOI](#)). These observations suggest that both neuron types may have evolved diverse structural features to support specialized sensory functions. One possibility is that this diversity expands the range of sensory surface area among homotypic ORNs. In the ab1C population, surface area spans 1.78 folds, ranging from 30.67 μm² to 54.75 μm². In the ab1D population, the span is 2.21 folds, ranging from 11.77 μm² to 26.08 μm².

Future functional assays will be needed to determine whether the observed range of morphometric variation among homotypic *Drosophila* ORNs is sufficient to broaden their odor sensitivity, akin to the phenomenon observed in rodent homotypic ORNs, where varying odor sensitivities correlates with different cilia lengths (Challis et al., 2015 [DOI](#); Grosmaître et al., 2006 [DOI](#)). It would also be valuable to investigate whether the dendritic heterogeneity observed among homotypic adult ORNs represent dynamic morphological plasticity regulated in an activity-dependent manner, which may provide a structural mechanism for olfactory adaptation. These exciting open questions highlight the need for technological advances. In particular, linking nanoscale morphology to neuronal function would offer critical insights. However, this remains

technically challenging, as it would require SBEM imaging of neurons previously recorded via single-sensillum electrophysiology. At present, no dye-labeling method is compatible with both single-sensillum recording and the cryofixation and sample preparation required for SBEM.

Alternatively, this morphological heterogeneity may have little functional significance and instead arise as a byproduct of the non-binary, graded nature of the molecular mechanisms underlying ORN-specific dendritic morphogenesis. For instance, in *C. elegans* amphid sensilla, the distinctive sensory cilia morphologies observed in AWA (tree-like branches), AWB (flattened, narrow protrusions), and AWC (flattened, wing-shaped) olfactory sensory neurons (Doroquez et al., 2014 [↗](#); Perkins et al., 1986 [↗](#)) are the result of differential expression levels of a single immunoglobulin domain membrane protein, OIG-8 (Howell & Hobert, 2017 [↗](#); Wallace & Shaham, 2017 [↗](#)). If a similar mechanism determines the dendritic morphologies of ab1C and ab1D neurons, the variability in morphogen expression levels—within the deterministic threshold specific to each neuronal type—could potentially explain the observed heterogeneity in dendritic morphology within the same ORN type.

Our morphometric analysis provides a wealth of data that paves the way for biologically realistic modeling in future studies. For example, one intriguing question involves the trade-offs of having extremely fine dendritic branches, such as the slender ab1D distal dendrite with a volume of 0.06 μm^3 . Smaller volumes increase the SA/V ratio, and amplify localized changes in ion concentrations, potentially enhancing signal conduction. For instance, in vertebrate olfactory cilia with an estimated volume of 0.5 μm^3 , a 1 pA Na^+ influx can cause a significant ion flux of ~ 20 mM/s. However, a sharp Na^+ surge can inhibit Ca^{2+} clearance via the $\text{Na}^+/\text{Ca}^{2+}/\text{K}^+$ exchanger, hampering response termination. Additionally, it causes a rapid rise in osmotic pressure, increasing the risk of ciliary swelling and damage. This constraint likely explains why vertebrate ORNs rely on Cl $^-$ efflux rather than Na $^+$ influx as the primary transduction current (Reisert & Reingruber, 2019 [↗](#)). In contrast, *Drosophila* ORNs lack reported Cl $^-$ efflux currents and instead rely on cation influx for olfactory transduction (Benton, 2022 [↗](#)). Modeling the biophysical effects of this influx on fine outer dendrites would be valuable. Future studies should also compare the biophysical implications of dendritic flattening versus branching and explore the functional impact of the “dendrite-within-dendrite” phenomenon in ab1C. How does this unique structure affect odor detection and ephaptic coupling among neighboring neurons? These questions highlight exciting avenues for further research.

Materials and methods

Animals

D. melanogaster flies were reared on standard cornmeal food containing molasses at 25°C, $\sim 60\%$ humidity under a 12-h light/dark cycle in an incubator. To facilitate comparison with our published ORN morphometrics data (Nava Gonzales et al., 2021 [↗](#)), flies of the same genotype (*Or7a-GAL4>10xUAS-myc-APEX2-Orco*) were used to generate the antennal volume. The fly stock numbers are BDSC 91810 for *Or7a-GAL4* (Lin et al., 2015 [↗](#)), and BDSC 79214 for *10xUAS-myc-APEX2-Orco* (Tsang et al., 2018 [↗](#)). However, APEX2-mediated DAB staining was not applied in this study to label specific neurons. Experimental flies were collected upon eclosion, separated by sex, and co-housed in groups of ten. Female flies aged 6–8 days were used for experiments.

Antenna dissection was performed as follows. A fly was first wedged into the narrow end of a truncated plastic 200- μl pipette tip to expose the antenna, which was subsequently stabilized between a tapered glass microcapillary and a coverslip covered with double-sided tape. To facilitate solution exchange during sample preparation, a sharp glass electrode was used to

puncture the lateral side of the antenna, a region with a low density of the ab1 sensillum type. Finally, the third antennal segment was severed from the fly's head by pinching the second segment with fine forceps.

Tissue preparation and SBEM volume acquisition

The SBEM volume of the *D. melanogaster* antenna was generated following the CryoChem protocol (Tsang et al., 2018 [DOI](#)). Briefly, dissected antennae were immediately subject to high-pressure freezing in a solution of 0.15 M sodium cacodylate and 20% BSA using a high-pressure freezing machine (Bal-Tec HPM 010). The frozen samples were then transferred to a freeze-substitution solution containing 0.2% glutaraldehyde (#18426, Ted Pella, USA), 0.1% uranyl acetate (Electron Microscopy Sciences, USA), and 1% water in acetone (#AC326800010, ACROS Organics, USA) in a liquid nitrogen bath. Freeze substitution was performed in a Leica EM AFS2 device held at -90 °C for 58 hours, from -90 °C to -60 °C for 15 hours, at -60 °C for 15 hours, from -60 °C to -30 °C for 15 hours, and then at -30 °C for 15 hours. During the final hour at -30°C, samples were washed three times (20 minutes each) in an acetone solution containing 0.2% glutaraldehyde and 1% water, before being transferred to ice for one hour prior to rehydration (see below)

The cryofixed samples were gradually rehydrated in a series of nine rehydration solutions, each for 10 minutes on ice.

1. 5% water, 0.2% glutaraldehyde in acetone
2. 10% water, 0.2% glutaraldehyde in acetone
3. 20% water, 0.2% glutaraldehyde in acetone
4. 30% water, 0.2% glutaraldehyde in acetone
5. 50% 0.1M HEPES (Gibco, Taiwan), 0.2% glutaraldehyde in acetone
6. 70%, 0.1M HEPES, 0.2% glutaraldehyde in acetone
7. 0.1M HEPES
8. 0.1M sodium cacodylate with 100 mM glycine
9. 0.1M sodium cacodylate

After rehydration, the samples were subject to *en bloc* heavy metal staining in a solution of 2% OsO₄, 1.5% potassium ferrocyanide, and 2 mM CaCl₂ in 0.1 M sodium cacodylate for one hour at room temperature. Then samples were washed five times with water (5 minutes per wash) before being incubated in 0.5% thiocarbohydrazide (Electron Microscopy Sciences, USA) for 30 minutes at room temperature. Following another series of water washes, the samples were incubated in 2% OsO₄ for 30 minutes at room temperature. After a final rinse with water, the samples were transferred to 2% aqueous uranyl acetate (filtered with 0.22 µm filter) at 4°C overnight. The samples were then washed with water and then subjected to the dehydration steps described below.

Dehydration was performed in six consecutive 10-minute steps: 70% ethanol, 90% ethanol, 100% ethanol (twice), followed by 100% acetone (twice). All ethanol steps were conducted on ice. The first acetone step used ice-cold acetone, while the second was performed with acetone at room temperature.

Resin infiltration was carried out in a 1:1 solution of Durcupan ACM resin and acetone and incubated overnight on a shaker. The samples were then transferred in fresh 100% Durcupan ACM resin twice, with a 6–7 hour interval between transfers. During incubation in 100% resin, the samples were placed in a vacuum chamber on a rocker to facilitate the evaporation of residual acetone. After an overnight incubation in 100% resin, the samples were embedded in fresh resin and polymerized at 60°C for at least two days. The composition of Durcupan ACM resin (Sigma Aldrich) was 11.4 g component A, 10 g component B, 0.3 g component C, and 0.1 g component D.

Microcomputed X-ray tomography was used to determine the position and proper orientation of the resin-embedded specimens. Samples were mounted on aluminum pins with conductive silver epoxy (Ted Pella) and sputter coated with gold-palladium for SBEM imaging with a Gemini SEM 300 (Zeiss) equipped with a Gatan 3View 2XP microtome system and the OnPoint backscatter detector.

The antennal SBEM volume was acquired at 2.5kV using a 30- μ m aperture, with the electron gun set to analytic mode and the beam operating in high-current mode. Nitrogen gas was used for focal charge compensation to reduce charging artefacts. Imaging was performed with a dwell time of 1 μ s, a pixel size of 5nm, and a Z-step of 40nm. The X and Y pixel numbers were 1909 and 2061, respectively, and there were a total of 2571 Z slices. After data collection, the images were converted to MRC format, and rigid alignment of the image slices was performed using cross-correlation in the IMOD image processing package (<https://bio3d.colorado.edu/imod/>). The SBEM volume is available in the Cell Image Library (<https://www.cellimagelibrary.org/>) with the accession numbers CIL:57519.

Image segmentation

In the fly antenna volume, the ab1 sensilla were identified based on their characteristic size and the presence of four neurons, as ab1 is the only large basiconic sensillum type housing four ORNs (Nava Gonzales et al., 2021; Shanbhag et al., 1999). Manual segmentation was conducted using IMOD's drawing tools by placing closed contours around the structures of interest in serial sections (Kremer et al., 1996). The sensillum cuticle, ORN soma, and inner and outer dendritic segments were saved as distinct objects to facilitate morphometric measurement of individual structures. The ciliary constriction was used to define the boundary between the inner and outer dendrites (Shanbhag et al., 2000).

For the subset of ab1C neurons featuring fully curled, tube-like outer dendrites, the enclosed dendritic regions were segmented into two separate objects: the hollow inner region and the membrane-bound outer region. Subsequently, all segmented objects were “meshed” to connect adjacent contours, creating continuous 3D structures. Detailed information about “imodmesh” and IMOD's drawing tools is available in the IMOD user guide (<https://bio3d.colorado.edu/imod/doc/man/imodmesh.html>; <https://bio3d.colorado.edu/imod/doc/3dmodHelp/pluginhelp/drawingtools.html>). All 3D neuron models generated in this study are available in the NIH 3D repository (<https://3d.nih.gov/>) under the entry ID: 3DPX-021684.

SBEM image post-processing

For representative SBEM images, image quality was enhanced using the DenoiseEM plug-in for ImageJ, which offers multiple denoising algorithm options. Briefly, TIFF images were first loaded into ImageJ and converted to a 16-bit file format. Multiple regions of interest within the sensillar lumen were sampled to train the denoising algorithms, and the optimal algorithm was selected based on the best signal-to-blur ratio or overall image quality. For the SBEM images presented in this study, the Gaussian algorithm was most frequently used. To further enhance the visibility of dendritic branches, the contrast and brightness of the denoised images were adjusted in ImageJ. The final images were then converted back to RGB format and exported as TIFF files. Detailed information about DenoiseEM is available in the DenoiseEM plug-in page (<https://bioimagingcore.be/DenoisEM/>).

3D model videos

Movies for each 3D ORN model were created using IMOD's “Movie Montage Control” and “Movie Sequence Dialog” functions. Briefly, “Movie Montage Control” allows users to manipulate 3D models and record these manipulations or scenes as a series of images using the “Set Start” and “Set End” controls. IMOD then interpolates a user-specified number of frames between the two

views to create a smooth transition. This workflow supports creating multiple scenes (e.g., zooming in/out, rotations, or displaying image planes) as a sequence using the “Movie Sequence Dialog”. Each scene in the sequence was exported as TIFF images. These images were then compiled into an image sequence in QuickTime and saved as a .mov file at 60 frames per second. Detailed information about IMOD’s movie and montage controls is available in the IMOD user guide (<https://bio3d.colorado.edu/imod/doc/3dmodHelp/modelMovie.html> )

Skeletonization

To visualize dendritic branching patterns, the 3D models of ORN dendrites in MOD format were first converted to VRML2 files using the command “imod2vrml2” in IMOD. The VRML2 files were then imported into Amira (2020.2 version; ThermoFisher Scientific, USA) and converted into a binary mesh, with the 3D model area colored in white and the background in black. The AutoSkeleton module in Amira was used to identify the center of mass within each mesh region and generate a 2D skeleton. The skeletons were then manually edited using the “Filament editor” in Amira by overlaying them with ORN 3D models to correct errors such as extra loops or branches.

These skeletons, in SWC format, were imported into neuTube software (<https://www.neutraceing.com/> ) , where dendritic branches were manually spread onto a 2D plane. Briefly, a primary branch and all its downstream branches were first selected to allow all the branches to be edited and moved as a group. This process was repeated for secondary, tertiary and higher-order branches until overlap between branches was minimized.

Morphometric analysis

For morphometric analysis, the sensillum cuticle, ORN soma, inner dendrite, and the proximal and distal outer dendritic segments were analyzed as separate objects.

Surface area and volume

The morphometric values were extracted from individual objects using the “imodinfo” function in IMOD. Detailed information about “imodinfo” is available in the IMOD User’s Guide (<https://bio3d.colorado.edu/imod/doc/man/imodinfo.html> ) . Among the three volume measurement options in “imodinfo”, the “Volume Inside Mesh” option was selected to measure ORN volumes. For enclosed tube-like structures in the ab1C outer dendrites, volume was calculated by subtracting the inner object volume from the outer object volume, while surface area was determined by summing the surface areas of the inner and outer objects. For the branched outer dendrites of ab1D, the total volume and surface area were obtained by summing the values from individual branches.

Absolute length

To calculate the absolute lengths of individual objects, these structures were first skeletonized using Amira. The resulting SWC files were imported into R, where pixel coordinates were scaled to micrometers using scaling factors derived from the ‘imodinfo’ command. The length of each component was then calculated using the Pythagorean theorem.

Relative position to sensillum cuticle

To calculate the relative positions of dendritic branching, flattening, or terminal points with respect to the cuticle, the previously mentioned SWC skeletons were first imported into Python. Branching points and dendritic termini were extracted using the PyNeuroML package (<https://pyneuroml.readthedocs.io/en/development/> ) . Each point of interest (e.g., ciliary constriction, branching or flattening point, and dendritic terminus) was projected onto the nearest point on the cuticle skeleton. The cuticle proportion of a point was defined as its relative position along the cuticle, scaled from 0 (cuticle base) to 1 (cuticle tip). For points below the cuticle base, the bottom

segment of the cuticle skeleton was extrapolated, and the point was projected onto this segment. In such cases, the cuticle proportion was scaled from 0 (at the cuticle base) to negative infinity, with -1 representing one cuticle length down the extrapolated segment.

Total number of dendrites in ab1 sensilla

The dendritic branch count for each ab1 sensillum, as shown in **Figure 1B**, was estimated using an image slice taken at approximately the midpoint of the sensillum cuticle. This image plane was rotated to a perpendicular orientation relative to the long axis of the cuticle using IMOD's slicer tool. In this orientation, the images were exported in TIFF format, denoised, and branches were manually counted. The sensillum cross-sectional area at the midpoint was measured by outlining the sensillar lumen with a contour and calculating its area using the 'imodinfo' command in IMOD. For more information about IMOD's slicer tool, refer to the IMOD user guide (<https://bio3d.colorado.edu/imod/doc/3dmodHelp/slicer.html>).

Statistics

All values were presented as mean $\pm$ standard deviation (SD). Paired two-tailed *t*-tests were used for morphometric comparisons between neighboring ORNs within the same sensillum. For comparisons between non-neighboring neurons, rank sum tests or unpaired two-tailed *t*-tests (if the Shapiro-Wilk normality test was passed) were applied. A *P* value of <0.05 was considered statistically significant.

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Additional files

Figure 1—figure supplement 1. *Drosophila* antenna and olfactory sensillum. (A) Schematic of the *Drosophila* antenna, which houses hundreds of olfactory sensilla on the surface of its third segment. (B) Schematic of an olfactory sensillum, illustrating the multiporous cuticle and compartmentalized olfactory receptor neurons (ORNs). Each ORN consists of an outer dendrite, inner dendrite, soma, and axon. The inner and outer dendrites are separated by the ciliary constriction.

Figure 2—figure supplement 1. Partially reconstructed ab1 sensillum with four neurons. (A) 3D model and SBEM images of a partially reconstructed ab1 sensillum from another SBEM volume. ORNs are pseudocolored to indicate neuronal identities: ab1A (blue), ab1B (orange), ab1C (yellow) and ab1D (white). Dashed lines mark the positions of corresponding SBEM images on the right: (1) highly branched outer dendrites; (2) proximal outer dendrites; and (3) unbranched segments of outer dendrites. The somata and some inner dendrites were truncated during SBEM image acquisition. (B–E) 3D models and corresponding 2D projections of the outer dendritic branches of ab1A (B), ab1B (C), ab1C (D), and ab1D (E). Scale bars: 2 μ m for 3D models and 1 μ m for SBEM images.

Video 1. 3D dendritic models of ab1A, B, C, and D neurons. Color codes are as indicated in Figure 2.

Video 2. 3D models representing distinct morphological types of ab1C dendrites.

Video 3. 3D models of auxiliary cells in the ab1 sensillum.

Source data file [↗](#)

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Reviewer #1 (Public review):

The authors of this study use electron microscopy and 3D reconstruction techniques to study the morphology of distinct classes of *Drosophila* sensory neurons *across many neurons of the same class.* This is a comprehensive study attempting to look at nearly all the sensory neurons across multiple sensilla in the same animal to determine a) how much morphological variability exists between and within neurons of different and similar sensory classes and b) identify dendritic features that may have evolved to support particular sensory functions. This study builds upon the authors' previous work which allowed them to identify and distinguish sensory neuron subtypes in the EM volumes without additional staining so that reconstructed neurons could reliably be placed in the appropriate class. This work is unique in looking at a large number of individual neurons of the same class to determine what is consistent and what is variable about their class-specific morphologies.

This means that in addition to providing specific structural information about these particular cells, the authors explore broader questions of how much morphological diversity

exists between sensory neurons of the same class. This then informs our conceptualization about how different dendritic morphologies might affect specific sensory and physiological properties of neurons.

The authors found that CO₂ sensing neurons have an unusual, sheet-like morphology in contrast to the thin branches of odor-sensing neurons. They show that this morphology greatly increases the surface area to volume ratio above what could be achieved by modest branching of thin dendrites, and posit that this might be important for their sensory function, though this was not directly tested in their study due to technical limitations. The study is mainly descriptive in nature, but thorough, and provides a nice jumping off point for future functional studies. One interesting future analysis could be to examine all four cell types within a single sensilla together to see if there are any general correlations that could reveal insights about how morphology is determined and relative contributions of intrinsic mechanisms vs interactions with neighboring cells. For example, if higher-than-average branching in one cell type correlated with higher-than-average branching in another type when within the same sensilla, it might suggest differential amounts of extracellular growth or branching cues within a given sensillum drive any heterogeneity observed within a class across sensilla. Conversely, if higher branching in one cell type consistently leads to reduced length or branching of the other neurons within its sensillum, this might point to dendrite-dendrite interactions between cells undergoing competitive or repulsive interactions to define territories within each sensillum as a major determinant of the variability.

Strengths:

This work provides a thorough morphometric analysis of the neurons of the *majority of all ab1 sensilla* across a single antenna. The authors use this analysis to 1) characterize the unique dendritic architecture of ab1C neurons relative to other ORNs including ab1D and 2) provide evidence of substantial morphological diversity even within a single subclass of neuron.

Weaknesses:

This is primarily a descriptive paper due to technical limitations since it is not currently technically feasible to determine individual ORN response properties and tie them to identified neurons with detailed EM-based ultrastructural analyses, nor to predictably alter dendritic morphology of these cells to directly test how different morphologies affect sensory function. However, the quantitative descriptive findings presented here will shape these future questions and are necessary for any such future work.

<https://doi.org/10.7554/eLife.106389.2.sa3>

Reviewer #2 (Public review):

Summary:

The manuscript employs serial block-face electron microscopy (SBEM) and cryofixation to obtain high-resolution, three-dimensional reconstructions of *Drosophila* antennal sensilla containing olfactory receptor neurons (ORNs) that detect CO₂. This method has been used previously by the same lab in Gonzales et. al, 2021. (<https://elifesciences.org/articles/69896>), and Zhang et. al, 2019 Nature Communications. The previous study by Zhang also correlated morphometric measurements from SBEM with asymmetric ephaptic activity for paired neurons using electrophysiology across multiple olfactory sensilla. This manuscript applies the same SBEM method to now characterize the ab1 sensillum which houses the ab1C, CO₂ detecting neuron, but stops short of integration neuronal activity with structural variability.

The SBEM-based morphometric studies do however significantly advance preliminary observations from older two-dimensional TEM-based reports. Previous images of the putative CO₂ neuron in *Drosophila* (Shanbhag et al., 1999) and in mosquitoes (McIver and Siemicki, 1975; Lu et al, 2007) reported that the dendritic architecture of the CO₂ neuron was somewhat different (circular and flattened, lamellated) from other olfactory neurons in the antenna of insects. In this study, the authors confirm this different morphology but also classify it into distinct subtypes (loosely curled, fully curled, split, and mixed).

Strengths:

The study makes a convincing case that ab1C neurons exhibit a unique, dendritic morphology unlike the canonical cylindrical dendrites found in ab1D neurons. This observation extends previous qualitative TEM findings by not only confirming the presence of flattened lamellae in CO₂ neurons but also quantifying key morphometrics such as dendritic length, surface area, and volume, and calculating surface area-to-volume ratios. The enhanced ratios observed in the flattened segments are speculated to be linked to potential advantages in receptor distribution (e.g., Gr21a/Gr63a) and efficient signal propagation.

Weaknesses:

Although this quantitative approach is very robust compared to earlier reports, interpretations are somewhat limited by the absence of direct electrophysiological data to confirm whether ultrastructural differences translate into altered neuronal function. The biggest question remains unanswered: whether structural variation observed in the ab1C dendrites by SBEM have an electrophysiological functional relevance?

Surveys of ab1 sensillum with single-sensillum recordings (even a few from multiple *Drosophila* antenna) as they have done for ab2s and others in the past, would have measured spontaneous activity, spike amplitude, and response to CO₂. This could have allowed for comparison of frequency of functional variation, if any, to structural variation and a discussion would therefore have strengthened the overall characterization. In the case of ab2 sensilla the authors find very little variance, could the ab1 also be the same? In the absence of this data, it becomes hard to speculate whether structural variation observed in the ab1C dendrites by SBEM have any functional relevance or whether they are simply random variations in dendrite development.

Additionally, artifacts could be a consideration, even though Cryofixation is superior to chemical fixation. Although this is hard to address, all types of fixations in TEMs cause some artifacts, as does serial sectioning. An understanding of the error rates for the SBEM method would have increased the confidence in the conclusions drawn. For example, what is the structural variation of SBEMs in the ab2 population, which shows very little electrophysiological variation? Can a comparison be done?

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Reviewer #3 (Public review):

Summary:

In the current manuscript entitled "Population-level morphological analysis of paired CO₂- and odor-sensing olfactory neurons in *D. melanogaster* via volume electron microscopy", Choy, Charara et al. use volume electron microscopy and neuron reconstruction to compare the dendritic morphology of ab1C and ab1D neurons of the *Drosophila* basiconic ab1 sensillum. They aim to investigate the degree of dendritic heterogeneity within a functional class of neurons using ab1C and ab1D, which they can identify due to the unique feature of

ab1 sensilla to house four neurons and the stereotypic location on the third antennal segment. This is a great use of volumetric electron imaging and neuron reconstruction to sample a population of neurons of the same type. Their data convincingly shows that there is dendritic heterogeneity in both investigated populations and their sample size is sufficient to strongly support this observation. This data proposes that the phenomenon of dendritic heterogeneity is common in the *Drosophila* olfactory system and will stimulate future investigations into the developmental origin, functional implications and potential adaptive advantage of this feature.

Moreover, the authors discovered that there is a difference between CO₂- and odour sensing neurons of which the first show a characteristic flattened and sheet-like structure not observed in other sensory neurons sampled in this and previous studies. They hypothesize that this unique dendritic organization which increases the surface area to volume ratio, might allow more efficient CO₂ sensing by housing higher numbers of CO₂ receptors. This is supported by previous attempts to express CO₂ sensors in olfactory sensory neurons which lack this dendritic morphology, resulting in lower CO₂ sensitivity compared to endogenous neurons.

Overall, this detailed morphological description of olfactory sensory neurons' dendrites convincingly shows heterogeneity in two neuron classes with potential functional impacts for odour sensing.

Strength:

The volumetric EM imaging and reconstruction approach offers unprecedented details in single cell morphology and compares dendrite heterogeneity across a great fraction of ab1 sensilla. The authors identify specific shapes for ab1C sensilla potentially linked to their unique function in CO₂ sensing.

Weaknesses:

While the morphological description is highly detailed, current methods prevent linking morphology to odour sensitivity or other properties of the neurons. Therefore, this study remains mainly descriptive and will require future work to link neuron structure and function.

<https://doi.org/10.7554/eLife.106389.2.sa1>

Author response:

The following is the authors' response to the original reviews

Reviewer #1 (Public Review):

*The authors of this study use electron microscopy and 3D reconstruction techniques to study the morphology of distinct classes of Drosophila sensory neurons *across many neurons of the same class.* This is a comprehensive study attempting to look at nearly all the sensory neurons across multiple sensilla to determine a) how much morphological variability exists between and within neurons of different and similar sensory classes, and 2) identify dendritic features that may have evolved to support particular sensory functions. This study builds upon the authors' previous work, which allowed them to identify and distinguish sensory neuron subtypes in the EM volumes without additional staining so that reconstructed neurons could reliably be placed in the appropriate class. This work is unique in looking at a large number of individual neurons of the same class to determine what is consistent and what is variable about their class-specific morphologies.*

This means that in addition to providing specific structural information about these particular cells, the authors explore broader questions of how much morphological diversity exists between sensory neurons of the same class and how different dendritic morphologies might affect sensory and physiological properties of neurons.

The authors found that CO₂-sensing neurons have an unusual, sheet-like morphology in contrast to the thin branches of odor-sensing neurons. They show that this morphology greatly increases the surface area to volume ratio above what could be achieved by modest branching of thin dendrites, and posit that this might be important for their sensory function, though this was not directly tested in their study. The study is mainly descriptive in nature, but thorough, and provides a nice jumping-off point for future functional studies. One interesting future analysis could be to examine all four cell types within a single sensilla together to see if there are any general correlations that could reveal insights about how morphology is determined and the relative contributions of intrinsic mechanisms vs interactions with neighboring cells. For example, if higher than average branching in one cell type correlated with higher than average branching in another type, if in the same sensilla. This might suggest higher extracellular growth or branching cues within a sensilla. Conversely, if higher branching in one cell type consistently leads to reduced length or branching in another, this might point to dendrite-dendrite interactions between cells undergoing competitive or repulsive interactions to define territories within each sensilla as a major determinant of the variability.

We thank the reviewer for the insightful comments and appreciation for our study.

Reviewer #2 (Public Review):

The manuscript employs serial block-face electron microscopy (SBEM) and cryofixation to obtain high-resolution, three-dimensional reconstructions of Drosophila antennal sensilla containing olfactory receptor neurons (ORNs) that detect CO₂. This method has been used previously by the same lab in Gonzales et. al, 2021. (<https://elifesciences.org/articles/69896>), which had provided an exemplary model by integrating high-resolution EM with electrophysiology and cell-type-specific labeling.

We thank the reviewer for expressing appreciation for our published study.

The previous study ended up correlating morphology with activity for multiple olfactory sensillar types. Compared to the 2021 study, this current manuscript appears somewhat incomplete and lacks integration with activity.

We thank the reviewer for their feedback. However, we would like to clarify that our previous study did not correlate morphology with activity to a greater extent than the current study. Both employed the same cryofixation, SBEM-based approach without recording odor-induced activity, but the focus of the current work is fundamentally different. While the previous study examined multiple sensillum types, the current study concentrates on a single sensillum type to address a distinct biological question regarding morphological heterogeneity. We appreciate the opportunity to clarify this distinction, and we hope that the revised manuscript more clearly conveys the unique scope and contributions of this study.

In fact older studies have also reported two-dimensional TEM images of the putative CO₂ neuron in Drosophila (Shanbhag et al., 1999) and in mosquitoes (McIver and Siemicki, 1975; Lu et al, 2007), and in these instances reported that the dendritic architecture of the CO₂ neuron was somewhat different (circular and flattened, lamellated) from other olfactory neurons.

We thank the reviewer for pointing this out. As noted in both the Introduction and Discussion sections, previous studies—including those cited by the reviewer—suggested that CO₂-sensing neurons may have a distinct dendritic morphology. However, those earlier studies lacked the means to definitively link the observed morphology to CO₂ neuron identity.

In contrast, our study assigns neuronal identity based on quantitative morphometric measurements, allowing us to confidently associate the unique dendritic architecture with CO₂ neurons. Furthermore, we extend previous observations by providing full 3D reconstructions and nanoscale morphometric analyses, offering a much more comprehensive and definitive characterization of these neurons. We believe this represents a significant advancement over earlier work.

The authors claim that this approach offers an artifact-minimized ultrastructural dataset compared to earlier. In this study, not only do they confirm this different morphology but also classify it into distinct subtypes (loosely curled, fully curled, split, and mixed). This detailed morphological categorization was not provided in prior studies (e.g., Shanbhag et al., 1999).

We thank the reviewer for acknowledging the significance of our study.

The authors would benefit from providing quantitative thresholds or objective metrics to improve reproducibility and to clarify whether these structural distinctions correlate with distinct functional roles.

We thank the reviewer for raising this point. However, we would like to clarify that assigning neurons to strict morphological subtypes was not the primary aim of our study. In practice, dendritic architectures can be highly complex, with individual neurons often displaying features characteristic of multiple subtypes. This is precisely why we included a “mixed” subtype category—to acknowledge and capture this morphological heterogeneity rather than impose rigid classification boundaries.

Our intent in defining subtypes was not to imply discrete functional classes, but rather to highlight the range of morphological variation observed across ab1C neurons. While we agree that exploring potential correlations between structure and function is an important future direction, the current study focuses on characterizing this diversity using 3D reconstruction and morphometric analysis. We hope this clarifies the purpose and scope of our morphological categorization.

Strengths:

The study makes a convincing case that ab1C neurons exhibit a unique, flattened dendritic morphology unlike the cylindrical dendrites found in ab1D neurons. This observation extends previous qualitative TEM findings by not only confirming the presence of flattened lamellae in CO₂ neurons but also quantifying key morphometrics such as dendritic length, surface area, and volume, and calculating surface area-to-volume ratios. The enhanced ratios observed in the flattened segments are speculated to be linked to potential advantages in receptor distribution (e.g., Gr21a/Gr63a) and efficient signal propagation.

We thank the reviewer for appreciating the significance of our current study.

Weaknesses:

While the manuscript offers valuable ultrastructural insights and reveals previously unappreciated heterogeneity among CO₂-sensing neurons, several issues warrant further

investigation in addition to the points made above.

(1) Although this quantitative approach is robust compared to earlier descriptive reports, its impact is somewhat limited by the absence of direct electrophysiological data to confirm that ultrastructural differences translate into altered neuronal function. A direct comparison or discussion of how the present findings align with the functional data obtained from electrophysiology would strengthen the overall argument.

We thank the reviewer for this comment. We would like to clarify, however, that our study does not claim that the observed morphological heterogeneity necessarily leads to functional diversity. Rather, we consider this as a possible implication and discuss it as a potential question for future research. This idea is raised only in the Discussion section, and we are carefully not to present functional diversity as a conclusion of our study. Nonetheless, we have reviewed the relevant paragraph to ensure the language remains cautious and does not overstate our interpretation.

We also acknowledge the significance of directly linking ultrastructural features to neuronal function through electrophysiological recordings. However, at present, it is technically challenging to correlate the nanoscale morphology of individual ORNs with their functional activity, as this would require volume EM imaging of the very same neurons that were recorded via electrophysiology. Currently, there is no dye-labeling method compatible with single-sensillum recording and SBEM sample preparation that allows for unambiguous identification and segmentation of recorded ORNs at the necessary ultrastructural resolution.

To acknowledge this important limitation, we have added a paragraph in the Discussion section, as suggested, to clarify the current technical barriers and to highlight this as a promising direction for future methodological advances.

(2) Clarifying the criteria for dendritic subtype classification with quantitative parameters would enhance reproducibility and interpretability. Moreover, incorporating electrophysiological recordings from ab1C neurons would provide compelling evidence linking structure and function, and mapping key receptor proteins through immunolabeling could directly correlate receptor distribution with the observed morphological diversity.

Please see our response to the comment regarding the technical limitations of directly correlating ultrastructure with electrophysiological data.

In addition, we would like to address the suggestion of using immunolabeling to map receptor distribution in relation to the 3D EM models. Currently, antibodies against Gr21a or Gr63a (the receptors expressed in ab1C neurons) are not available. Even if such antibodies were available, immunogold labeling for electron microscopy requires harsh detergent treatment to increase antibody permeability, damaging morphological integrity. These treatments would compromise the very morphological detail that our study aims to capture and quantify.

(3) Even though Cryofixation is claimed to be superior to chemical fixation for generating fewer artifacts, authors need to confirm independently the variation observed in the CO2 neuron morphologies across populations. All types of fixation in TEMs cause some artifacts, as does serial sectioning. Without understanding the error rates or without independent validation with another method, it is hard to have confidence in the conclusions drawn by the authors of the paper.

We thank the reviewer for raising concerns regarding potential artifacts in morphological analyses. However, we would like to clarify that cryofixation is widely regarded as a gold

standard for ultrastructural preservation and minimizing fixation-induced artifacts, as supported by extensive literature. This is why we adopted high-pressure freezing and freeze substitution in our study.

We have also published a separate methods paper (Tsang et al., eLife, 2018) directly comparing our cryofixation-based protocol with conventional chemical fixation, demonstrating substantial improvements in morphological preservation. This provides strong empirical support for the reliability of our approach.

Regarding the suggestion to validate observed morphological variation across populations: we note that determining the presence of artifacts requires a known ground truth, which is inherently unavailable as we could not measure the morphometrics of fly olfactory receptor neurons in their native state. In the absence of such a benchmark, we have instead prioritized using the best-available preparation methods and high-resolution imaging to ensure structural integrity.

Addressing these concerns and integrating additional experiments would significantly bolster the manuscript's completeness and advancement.

We appreciate the reviewer's feedback. As discussed in our responses to the specific comments above, certain suggested experiments are currently limited by technical constraints, particularly in the context of high-resolution volume EM for insect tissues enclosed in cuticles.

Nevertheless, we have carefully addressed the reviewer's concerns to the fullest extent possible within the scope of this study. We have revised the manuscript to clarify methodological limitations, added new explanatory content where appropriate, and ensured that our interpretations remain well grounded in the data. We hope these revisions strengthen the clarity and completeness of the manuscript.

Reviewer #3 (Public Review):

*In the current manuscript entitled "Population-level morphological analysis of paired CO₂- and odor-sensing olfactory neurons in *D. melanogaster* via volume electron microscopy", Choy, Charara et al. use volume electron microscopy and sensillum. They aim to investigate the degree of dendritic heterogeneity within a functional class of neurons using *ab1C* and *ab1D*, which they can identify due to the unique feature of *ab1* sensilla to house four neurons and the stereotypic location on the third antennal segment. This is a great use of volumetric electron imaging and neuron reconstruction to sample a population of neurons of the same type. Their data convincingly shows that there is dendritic heterogeneity in both investigated populations, and their sample size is sufficient to strongly support this observation. This data proposes that the phenomenon of dendritic heterogeneity is common in the *Drosophila* olfactory system and will stimulate future investigations into the developmental origin, functional implications, and potential adaptive advantage of this feature.*

Moreover, the authors discovered that there is a difference between CO₂- and odour-sensing neurons of which the first show a characteristic flattened and sheet-like structure not observed in other sensory neurons sampled in this and previous studies. They hypothesize that this unique dendritic organization, which increases the surface area to volume ratio, might allow more efficient CO₂ sensing by housing higher numbers of CO₂ receptors. This is supported by previous attempts to express CO₂ sensors in olfactory sensory neurons, which lack this dendritic morphology, resulting in lower CO₂ sensitivity compared to endogenous neurons.

Overall, this detailed morphological description of olfactory sensory neurons' dendrites convincingly shows heterogeneity in two neuron classes with potential functional impacts for odour sensing.

Strength:

The volumetric EM imaging and reconstruction approach offers unprecedented details in single cell morphology and compares dendrite heterogeneity across a great fraction of ab1 sensilla. The authors identify specific shapes for ab1C sensilla potentially linked to their unique function in CO2 sensing.

We thank the reviewer for the insightful comments and appreciation for our study.

Weaknesses:

While the morphological description is highly detailed, no attempts are made to link this to odour sensitivity or other properties of the neurons. It would have been exciting to see how altered morphology impacts physiology in these olfactory sensory cells.

We agree that linking morphological variation to physiological properties, such as odor sensitivity, would be a highly valuable direction for future research. However, the aim of the current study is to provide an in-depth nanoscale characterization based on a substantial proportion of ab1 sensilla, highlighting morphological heterogeneity among homotypic ORNs.

At present, it is technically challenging to correlate the nanoscale morphology of individual ORNs with their physiological responses, as this would require volume EM imaging of the exact neurons recorded via single-sensillum electrophysiology. Currently, no dye-labeling method exists that is compatible with both single-sensillum recording and the stringent requirements of SBEM sample preparation to allow for unambiguous identification and segmentation of recorded ORNs.

To acknowledge this important limitation, we have added a paragraph in the *Discussion* section clarifying the current technical barriers and highlighting this as a promising area for future methodological development. Please also see our responses to the reviewer's 4th comment below, where we present preliminary experiments examining whether odor sensitivity varies among homotypic ORNs.

(Please see the following pages for additional responses to the reviewers' specific comments. These responses are not intended for publication.)

Reviewer #1 (Recommendations for the authors):

As this is mainly a descriptive paper I have no suggestions for additional experiments.

Minor Text Suggestions:

(1) The authors might want to include a better description/definition of the fly antennae, olfactory sensilla and their basic structure/makeup, position of the sensory neurons and dendrites within, etc, in the introduction perhaps in cartoon form to help readers that are not familiar (i.e. non-Drosophila readers) with the terminology and basic organization can follow the paper more easily from the start.

We thank the reviewer for the helpful suggestion to broaden the appeal of our study to a wider readership. In response, we added a new introductory paragraph at the beginning of the *Results* section, along with illustrations in a new supplementary figure (Figure 1—figure supplement 1). The new paragraph reads as follows.

“The primary olfactory organ in *Drosophila* is the antenna, which contains hundreds of olfactory sensilla on the surface of its third segment (Figure 1—figure supplement 1A). Each sensillum typically encapsulates the outer dendrites of two to four ORNs. The outer dendrites are the sites where odorant receptors are expressed, enabling the detection of volatile chemicals. A small portion of the outer dendrites lies beneath the base of the sensillum cuticle. At the ciliary constriction, the outer dendrites connect to the inner dendritic segment, which then links to the soma of each ORN (Figure 1—figure supplement 1B).”

(2) In Figure 4D, the letter annotations above the graphs are not clearly defined anywhere that I could easily find. Please clarify with different symbols and/or in the figure legend so readers can easily comprehend the stats that are presented.

We thank the reviewer for raising this point. As suggested, in the revised Figure 4D legend, following the original sentence “Statistical significance is determined by Kruskal-Wallis one-way ANOVA on ranks and denoted by different letters”, we added “For example, labels “a” and “b” indicate a significant difference between groups ($P < 0.05$), whereas labels with identical or shared letters (e.g., “a” and “a”, “a,b” and “a”, or “a,b” and “b”) indicate no significant difference.”

Reviewer #3 (Recommendations for the authors):

There are several aspects that I would like the authors to consider to improve the current manuscript:

(1) Line 331: “Our analysis highlights how structural scaling in ab1D neurons achieves enhanced sensory capacity while maintaining the biophysical properties of dendrites”. This is a strong statement, and not shown by the authors. They speculate about this in the discussion, but I would like them to soften the language here.

We thank the reviewer for raising this point. As suggested, we have softened the language in the sentence in question. The revised version is as follows.

“Our analysis suggests that structural scaling in ab1D neurons may enhance sensory capacity while preserving the biophysical properties of dendrites.”

(2) The Supplementary material is not well presented and is not cited in the manuscript. It is not clear what the individual data files show, where they refer to, etc. Please provide clear labels of all data, cite them at the appropriate location in the manuscript, and make them more accessible to the reader. Also, there are two Videos mentioned in the manuscript that are not included in the submission.

We thank the reviewer for bringing this to our attention and apologize for the oversight. We appreciate the reviewer’s careful attention to the supplementary materials. We have addressed these issues accordingly: 1) all source data have been consolidated in to a single, clearly labeled Excel file to improve accessibility for readers; this file is now cited at the appropriate locations in the manuscript. 2) The supplementary videos mentioned in the manuscript have also been included in the re-submission.

(3) In Figure 1B, it is hard to recapitulate the increase in dendritic density in the presented pictures. Could the authors please highlight dendrites in the raw imaging files (e.g. by colour coding as done later in the manuscript). Also, it might be helpful to indicate the measured parameters visually in this Figure (e.g. volume, length, etc.).

We thank the reviewer for the helpful suggestion. As suggested, we have pseudocolored the dendrites in Figure 1B to enhance visual clarity.

As noted, the original legend stated that “the sensilla were arranged from left to right in order of increasing dendritic branch counts”. To improve clarity, we have now added the number of dendritic branches above each sensillum to make this information more explicit.

We hope these changes make the figure more accessible and informative for readers.

(4) Given the strength of the authors in in vivo physiology and single sensilla recordings, I would be very curious about how the described morphological heterogeneity is reflected in the response properties of ab1Cs and ab1Ds. Can the authors provide data (already existing from their lab) of these two neurons on response heterogeneity? I acknowledge that spike sorting can be very challenging in ab1s, but maybe it is possible to show the range of response sensitivities upon CO2 stimulation in ab1Cs? The authors speculate in the discussion and presented data will only be correlative - however I think it would strengthen the manuscript to have some link to physiology included.

We thank the reviewer for this insightful comment. We share the same curiosity about response variability among homotypic ORNs, including ab1C and ab1D. Ideally, this question could be addressed by recording from a large proportion of neurons of a given ORN type to assess the response variability within a single antenna. However, due to technical limitations, we are only able to reliably record from 3–4 ab1 sensilla per antennal preparation, representing approximately 8% of the total ab1 population.

Moreover, our recordings are typically limited to ab1 sensilla located on the posterior-medial side of the antenna, as this region provides the best accessibility for our recording electrode. This spatial constraint may limit our ability to sample the full morphological diversity of ab1C and ab1D neurons.

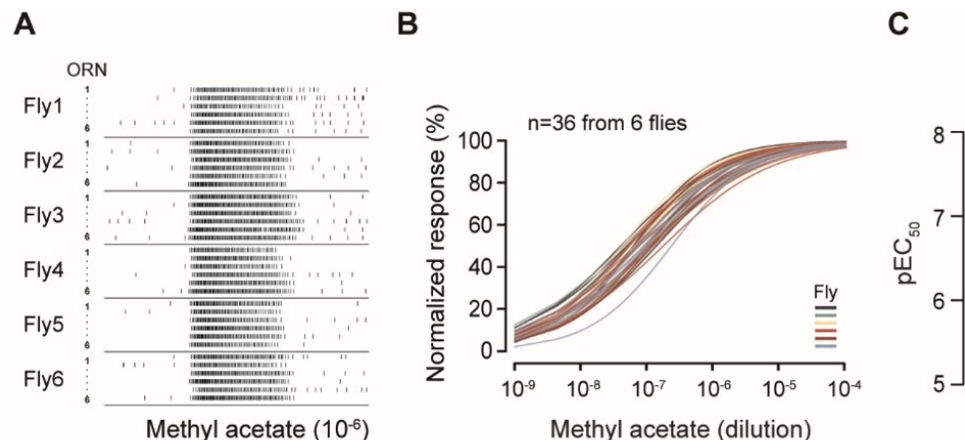
Given these limitations, it is technically challenging to rigorously assess physiological variability in ab1C and ab1D responses across the entire ab1 population. Nonetheless, we attempted to address this question using a different sensillum type where a larger proportion of the population is accessible to single-sensillum recording per antennal preparation. Specifically, we focused on ab2 sensilla in the following analysis because we can reliably record from 6 sensilla per antenna, representing approximately 25% of the total ab2 population.

In the preliminary data presented below, we recorded from 6 ab2A ORNs per antenna across a total of 6 flies. Spike analysis revealed that odor-evoked responses were consistent across individual ab2A neurons (Author response image 1A). When analyzing the dose-response curve for each ORN, we found no statistically significant differences in odor sensitivity, either among ORNs within the same antenna or across different flies (Author response image 1B; two-way ANOVA: $P > 0.99$ within antennae, $P > 0.99$ across flies). This is further supported by the closely clustered EC50 values (Author response image 1C). This result suggests that odor sensitivity is largely uniform among homotypic ab2A ORNs.

Author response image 1.

Homotypic ab2A ORNs display similar odorant sensitivity. (A) Single-sensillum recording. Raster plots of ab2A/Or59b ORN spike responses. Six ab2A ORNs from the same antenna were recorded per fly. Odor stimulus: methyl acetate (10-6). (B) Dose-response relationships of peak spike responses, normalized to the maximum response of the ORN to facilitate comparison of odor sensitivity. Each curve represents responses from a single ab2A ORN fitted with the Hill equation ($n=36$ ab2 sensilla from 6 flies). Responses recorded from the same antenna are indicated by the same color. Statistical comparisons between different ab2A ORNs from the

same antenna ($P > 0.99$) or across flies ($P > 0.99$) were performed by two-way ANOVA. (C) Quantification of individual pEC₅₀ values from (B), defined as $-\log EC_{50}$.



However, we are hesitant to include this result in the main manuscript for several reasons. First, it does not directly relate to the morphometric analysis of ab1C and ab1D neurons, which is the primary focus of our study. Second, while we were able to record from approximately 25% of the ab2 population, this level of coverage is still limited and potentially subject to sampling bias due to the spatial constraints of the antennal region accessible to the recording electrode.

At best, our data suggest limited variability in odor sensitivity among the recorded ab2A ORNs. However, we are cautious about generalizing this finding to the entire ab2 population. In light of these considerations, we hope the reviewer can appreciate the technical challenges inherent in addressing what may appear to be a straightforward question.

For these reasons, we have chosen to include this preliminary result in the response only, rather than in the main manuscript.

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